

Nathan (Kaiyan) Zhang

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Education

The University of British Columbia

Sep 2023 – May 2028

B.A., Combined Major in Economics and Statistics; Minor in Data Science

Cumulative average: 93.5/100

Selected coursework: Statistical Inference (100); Machine Learning and Data Mining (93); Causal Methods (98)

Advanced Econometrics (98); Statistical Learning (97); Time Series and Forecasting (95)

Experience

ESG Co-op, Capital Markets

Sep 2026 – Present

British Columbia Investment Management Corporation (BCI) / Vancouver, BC

- Support ESG risk and opportunity research; collect and analyze ESG data for investment research.

Research Assistant

May 2026 – Present

UBC Sauder School of Business | Supervisor: Allen Hu

- Reconciled 1,330 RavenPack publication values with 3,210,045 ProQuest articles (Jul 1861–May 2026); audited coverage, missingness, provenance, and field reliability.
- Built a collection and validation pipeline for an 85,259-ID financial-creator directory, preserving raw media, metadata, run manifests, evidence links, and reconciliation results.

Research Assistant

Apr 2026 – Present

UBC Sauder School of Business | Supervisors: J. Frank Li and Cheng Chen

- Built worker, task, and patent panels linking CPS, PSID, NLSY, O*NET, and PATSTAT as the sole RA on a faculty-led study of age-biased technological change; estimated wage and occupational sorting models.
- Harmonized 60+ O*NET releases and compared task text with about 56M PATSTAT records. Implemented share/log-share GMM, a 999-replication block bootstrap, and the R replication package.

Research Assistant

Jan 2025 – May 2026

Vancouver School of Economics at UBC | Supervisors: Jonathan Graves and Laura Nelson

- Built Python/R learning modules and Quarto/GitHub Actions publishing for econometrics, forecasting, NLP, and computer vision, supporting 1,000+ cumulative student users.
- Designed, hosted, and taught a computer vision workshop for social science research. Coauthored an image embedding notebook and a Programming Historian tutorial under site-maintainer review.

Selected Research

Memory Is the Difference between Two Futures: Memory Response Paths and Path-Structured Context Compilation

2026

Manuscript under review. With Jerrold Huang, designed a matched-future evaluation removing one memory at a time across four horizons in 180 independent ecologies. In a separate 144-item evaluation, PSCC exceeded A-Mem's exact-match accuracy by 6.94 percentage points (95% CI: 2.64 to 11.32), using 67.2% of its API tokens.

When Answers Surprise

2026

Manuscript under review. Built and deduplicated a panel of about 31,000 firm events (2018–2025) with Jerrold Huang, linking Capital IQ transcripts, IBES, CRSP, and historical S&P 500 membership. Tested whether management-answer novelty predicts returns beyond earnings surprise using time holdouts, learner comparisons, placebos, and timing checks.

Double Machine Learning for Payment-Friction Spending Effects

2025

Course research project. Used cross-fitted DoubleML on 5,583 respondents and 22,332 in-person person-days. Under selection on observables and overlap, the estimated conditional effect of lower-friction payment mixes on daily spending is about 5%, with heterogeneity by ticket size, income, and education.

Technical Skills

Programming & data: Python, R, SQL, Julia, DuckDB, Polars, Git/GitHub Actions, HPC/Slurm, Jupyter/Quarto

Modeling: causal inference/econometrics, panel/time series, machine learning, NLP, computer vision, scikit-learn, PyTorch, Transformers, fixest, statsmodels

Languages: Mandarin Chinese: native; English: academic working proficiency; Japanese: JLPT N3